

ANMOL GARG

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PROFILE

Network Engineer with CCNA and CCNP Enterprise training certifications (Network Bulls, Gurugram), holding hands-on experience in enterprise network design, routing and switching, and Cisco infrastructure. Complements core networking expertise with a strong foundation in AI/ML and software development — including deep learning, computer vision, and Python-based backend systems — gained through multiple research internships at IIT Mandi and independent projects. Combines a networking-first skill set with applied AI/ML capability, oriented toward roles in enterprise networking, NOC/network support, and network-adjacent automation.

SKILLS

Networking Fundamentals: OSI Model, TCP/IP, IPv4/IPv6 Addressing, Subnetting.

Routing Technologies: Static/Default Routing, RIP, EIGRP, OSPF, BGP, MPLS Fundamentals, HSRP.

Switching Technologies: VLANs, SVI, Trunking (802.1Q), VTP, STP/Rapid-PVST+, EtherChannel (LACP).

Network Services & Security: DHCP, DNS, NAT/PAT, SSH, ACLs.

Network Tools & Simulation: EVE-NG, Cisco Packet Tracer, Wireshark.

AI/ML & Data: TensorFlow, PyTorch, Keras, CNNs, Autoencoders, Object Detection, Image Segmentation, Transfer Learning, OpenCV.

Backend & Programming: Python, C++, Flask, FastAPI, REST APIs, Model Deployment, MySQL, PostgreSQL, Git/GitHub, Linux.

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Computer Networks, Operating Systems.

EXPERIENCE

Network Bulls — Network Engineer Trainee

Jan 2026 – Present

- Gained intensive hands-on experience with enterprise networking concepts through practical lab sessions and real-world network implementation scenarios.
- Built, tested, and troubleshooted complex network topologies while developing a structured approach to fault isolation and connectivity issue resolution.
- Performed network verification, monitoring, and troubleshooting using industry-standard methodologies and Cisco IOS diagnostic commands.

IIT Mandi — Research Intern (CAIR)

Jun – Jul 2025

- Developed a multi-stage data processing pipeline for railway infrastructure anomaly detection involving data preparation, validation, and automated workflow execution.
- Built ML pipelines using VGG-based segmentation and autoencoder reconstruction, achieving 98.66% accuracy with improved robustness to subtle and rare anomalies.
- Worked extensively in Linux-based environments, managing large datasets and optimizing processing workflows on compute-intensive, high-performance research infrastructure.

IIT Mandi — Research Intern

Jun – Jul 2024

- Conducted research on object detection using YOLO and RetinaNet with a VGG backbone for railway/track-related computer vision tasks.
- Gained hands-on experience in object recognition and computer vision workflows using PyTorch, TensorFlow, and OpenCV.

PROJECTS

Design and Simulation of Hierarchical Enterprise Network Architecture | EVE-NG, Cisco IOS, Cisco Routers & Switches [\[Link\]](#)

- Designed and implemented a scalable three-tier enterprise network architecture (Core, Distribution, Access) to support efficient communication across multiple departments.
- Configured VLANs, Inter-VLAN Routing, 802.1Q Trunking, VTP, NAT/PAT, DHCP, DNS, SSH v2, and Layer-2 security mechanisms for secure, centrally managed network communication.
- Implemented OSPF, HSRP, EtherChannel, Rapid-PVST+, and dual-ISP connectivity to ensure high availability, fault tolerance, and reliable network operation.
- Deployed centralized enterprise services (DHCP, DNS, NAT, web hosting) and performed network verification and troubleshooting using Cisco IOS commands and connectivity testing tools.

EIGRP + HSRP Dual-ISP Campus Lab | EIGRP, HSRP, NAT/PAT, Cisco IOS [\[Link\]](#)

- Built a dual-ISP campus topology using EIGRP as the dynamic routing protocol, with NAT/PAT configured at the edge routers for redundant internet connectivity.

- Configured HSRP-based first-hop gateway redundancy across multiple user VLANs on redundant distribution switches to remove single points of failure.
- Validated active/standby failover behavior and inter-VLAN reachability across the topology under simulated link-failure conditions.

Multi-Switch VTP, VLAN Segmentation & Spanning-Tree Optimization Lab / *VTP, VLANs, Rapid-PVST+, Layer-2 Security* [\[Link\]](#)

- Configured static trunk links with DTP disabled and set up VTP (SW3 as server, SW1/SW2 as clients) to propagate VLAN 10 (Sales), VLAN 20 (Accounts), and VLAN 100.
- Tuned Rapid-PVST+ so SW1 became the root bridge, adjusting bridge priority and port cost on SW2/SW3 to achieve deterministic root, designated, and blocking ports.
- Applied Layer-2 controls to restrict communication between Sales and Accounts VLANs and enabled fast convergence so alternate ports take over immediately on a topology change.

SmartCAP — AI-Powered Assistive Vision System for the Visually Impaired / *Python, Raspberry Pi, OpenCV, OCR, TTS, LiDAR, Face Recognition* [\[Link\]](#)

- Developed a Raspberry Pi-based assistive system integrating OCR-to-speech reading, LLM-based scene description, and face recognition for safety and personalized assistance.
- Used LiDAR for real-time distance estimation to obstacles, enabling safer navigation for visually impaired users.
- Optimized the speech-to-text/text-to-speech pipeline to reduce latency and minimize dependence on internet connectivity for faster, more reliable interaction.
- Winner, Project Gallery Competition — placed 1st among 16 teams at the college-level hackathon.

QWatch / SmartGUARD — Malicious QR Code & URL Detection System / *Python, Flask, JavaScript, scikit-learn* [\[Link\]](#)

- Built a hybrid ML model for detecting malicious QR codes and URLs, achieving 92% detection accuracy.
- Developed a Chrome extension integrated with a Flask backend for real-time URL and QR scanning.
- Integrated online learning for continuous model improvement and adaptability to emerging threats.

Brain Tumor Classification / *TensorFlow, OpenCV, Flask, HTML/CSS* [\[Link\]](#)

- Trained a MobileNet-based transfer learning model for MRI brain tumor classification.
- Built a Flask-based backend API and web UI for uploading MRI images and returning predictions.

RESEARCH WORK & PUBLICATIONS

Medical Image Classification Using Deep Learning [\[Publication\]](#)

- Implemented MobileNet, ResNet50, and VGG19 for multi-disease medical image classification, focused on improving accuracy and robustness for clinical diagnosis.
- Published and presented at Machine Intelligence for Research & Innovation (Maitri-2024), NIT Srinagar.

Leveraging Deep Learning for Lumpy Skin Disease Detection in Cattle [\[Publication\]](#)

- Developed a deep learning-based classification model to detect Lumpy Skin Disease (LSD) in cattle from medical images, implementing and optimizing VGG16, ResNet, and EfficientNet architectures.
- Published and presented at the Doctoral Symposium on Computational Intelligence.

EDUCATION

Jawaharlal Nehru Govt. Engineering College, Sundernagar, HP 2022 – 2026

B.Tech — Computer Science Engineering (AI & ML) | CGPA: 7.75

S.V.N.M.P. Sr. Sec. School, Kandroun, Bilaspur, HP 2021 – 2022

XII, HPBOSE | 88.6%

S.V.N.M.P. Sr. Sec. School, Kandroun, Bilaspur, HP 2019 – 2020

X, HPBOSE | 96.5%

CERTIFICATIONS & ACHIEVEMENTS

- **CCNP Enterprise Training** — Currently Pursuing, Network Bulls.
- **CCNA Training Certification** — Completed at Network Bulls, Gurugram (2026). [\[Certificate\]](#)
- **GATE DA 2025 Qualified** — Demonstrated strong analytical aptitude and problem-solving skills. [\[Score Card\]](#)
- **Deep Learning** | NPTEL (IIT Ropar) | 80% marks. [\[Certificate\]](#)
- **Remote Pilot Certificate (RPC)** — DGCA Approved, Adani Skill Development Centre.
- **Drone Bootcamp** — IIT Mandi — Exposure to embedded systems, sensor integration, and wireless communication concepts. [\[Certificate\]](#)
- **Workshop Certificate** — AI/ML using Python — IIT Mandi.
- **Winner, Project Gallery Competition** — College technical fest, 1st among 16 teams, for the SmartCAP AI-based solution. [\[Certificate\]](#)